James Wanjiku

Machine Learning Engineer and Team Lead | Data Science, Applied AI, MLOps
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SUMMARY

Machine learning engineer in my third year of building models and taking them to production, and team lead since July 2026. I define the problem and the metric with the people who own it, build and evaluate the model, deploy it, then monitor it and retrain it when it drifts. Work so far covers computer vision, NLP and LLM systems, including a duplicate-screening system for public funding applications that works much like fraud detection. Lead author of a peer-reviewed paper on LLM reliability (ICIST 2026).

SKILLS

Programming: Python, SQL, PostgreSQL, SQLAlchemy, TypeScript
Data science and machine learning: statistical modelling, supervised learning, feature engineering, model evaluation and validation, threshold and precision/recall tuning, experiment design, data drift checks and retraining, NLP, computer vision (detection, segmentation, tracking)
Generative AI and LLMs: LLM evaluation (LLM-as-judge, DeepEval), LangGraph, RAG, embeddings and vector search, prompt and agent design, guardrails
Libraries: PyTorch, Ultralytics YOLO, pandas, NumPy, FastAPI, Pydantic
MLOps and cloud: Google Cloud Platform, Docker, CI/CD with GitHub Actions, model monitoring and logging, TensorRT, NVIDIA Jetson, Redis, Git
Leadership: team lead, setting model targets, technical reviews, model documentation and validation reports

WORK EXPERIENCE

Team Lead, Machine Learning, AAI Labs (Lithuania, remote)

Jul 2026 - Present

- Lead the engineering team building onboard computer vision for aid-delivery drones on an EU-funded Eureka R&D project, working with industry and university partners. I set the model accuracy targets, plan the work, review results and still write code.
- Led the team through new simulated training data with varied lighting, an evaluation of non-ML trackers, a time-based safe-area check on the drone and a study of automatic labelling with a segmentation model.
- Retrained the ground hazard segmentation model on new open aerial data. Detection of water, the weakest class, improved sharply, and the other classes held steady.
- Compared model sizes for speed and accuracy on the edge device and kept the model that met the frame-rate budget with the best accuracy.
- Validated 3D trajectory reconstruction and wind estimation against simulation ground truth and wrote up the error bounds in a validation report.
- Set the team's retraining practice, so new models continue from saved checkpoints with the added data and are compared with the previous version on the same test set.
- Led the autonomous B2B voice calling platform from July to September 2026, and the team that delivered parallel calling, a simulation engine for testing calls without real phone lines, production hardening and the handover of its GPU services to the client's own cloud.

AI/ML Engineer, AAI Labs (Lithuania, remote)

Jul 2024 - Jun 2026

- Built an NLP system that compares large volumes of public funding applications using embeddings and vector search and flags likely duplicates for human review.
- Built a recommendation service on FastAPI and Google Cloud that suggests activities from historical records.
- On the voice calling platform (from June 2025), built much of the real-time conversation layer and a LangGraph state engine that blocks invalid conversation steps. Added automatic language detection, an assisted mode where a person can join a live call, latency alerts and separate build and deploy pipelines.
- Built the georeferencing and object tracking pipeline for a 5G-connected drone system. YOLO detection and multi-object tracking on live video, with each detection placed on GPS coordinates from drone telemetry. Presented to and accepted by an EU committee.
- Led a research study on how reliably LLMs follow instructions in multi-turn workflows and built its LLM-as-judge evaluation harness. Lead author of the resulting paper.
- Wrote the technical sections of EU research grant proposals.

Network Administrator, JayBee Internet
Aug 2023 - May 2024

- Maintained network infrastructure and supported users before moving into AI engineering.

EDUCATION

Bachelor of Science in Computer Science, University of Embu, Kenya, 2018 - 2023

CERTIFICATIONS

Machine Learning Specialization, Stanford University and DeepLearning.AI (Coursera)
Natural Language Processing Specialization, DeepLearning.AI (Coursera)
Mathematics for Machine Learning (Coursera)

PUBLICATIONS

J. Wanjiku, A. Zhurba, Yeabtsega. Towards Reliable Instruction-Following in LLM Multi-Turn Workflows: An Empirical Comparison of Prompting and Control Architectures. ICIST 2026, Kaunas University of Technology, Lithuania. Lead author and presenter.

PROJECTS

Taura, open-source Kikuyu-English machine translation. Cross-lingual embeddings with hybrid keyword and vector retrieval, served through FastAPI over 19,300 sentence pairs.
Nemo Voice Typing, offline speech recognition for dictation on Windows, running locally on CPU.